

Technical Report

AUTOSAR HLD Document Analysis Assistant using RAG with Citations

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1. Abstract

This project builds a RAG assistant for AUTOSAR High-Level Design documents. It answers engineer questions using only retrieved pages, cites the page for each answer, and refuses when the document has no answer. It is evaluated on a ground-truth question set.

2. Problem and objectives

HLD documents are long and unstructured, so finding a port, signal or dependency is slow and error-prone. Objectives: (a) searchable question answering over an HLD, (b) page citations on every answer, (c) safe refusal when the answer is missing, (d) measured accuracy.

3. Knowledge base

A synthetic 14-page Body Control Module (BCM) HLD (Input_Data/BCM_HLD_v1.0.pdf) generated by Code/generate_hld.py. It covers components, ports, CAN signals, functional flows, BSW dependencies, diagnostics and revision history, with text and tables. No real or confidential data is used.

4. System design

PDF -> pdfplumber text per page -> section chunks (max 800 characters, section title kept) -> BAAI/bge-small-en-v1.5 embeddings -> ChromaDB (cosine) -> top-4 retrieval -> llama3.2:3b (Ollama, temperature 0) with a strict prompt -> answer with [Page N] citations -> Streamlit UI.

Item	Value
LLM	llama3.2:3b via Ollama (local)
Embeddings	BAAI/bge-small-en-v1.5
Vector store	ChromaDB, persistent, cosine distance
Top-k	4
Refusal threshold	best-match distance above 0.335 gives 'Not found'
Prompt	Model_Prompts_Config/system_prompt.txt
UI	Streamlit

5. Evaluation method

30 questions with known answers and pages (Input_Data/test_questions.json): 28 answerable, 2 not in the document. Metrics: retrieval hit rate (expected page among retrieved pages), answer accuracy (all required keywords present), citation accuracy (expected page cited), and correct refusal rate on unanswerable questions. The evaluation was run in extractive mode (retrieval + best passage extraction) because Ollama was not available. The LLM generation step is implemented in rag.py but was not evaluated.

6. Results

Metric	Value
Answer mode used	extractive
Answerable questions	28
Retrieval hit rate	96.4 %
Answer accuracy	78.6 %
Citation accuracy	78.6 %
Correct refusal rate (unanswerable)	100.0 %

Questions answered incorrectly (from results.csv):

Q	Question	System answer (short)
4	Which team owns LightingController?	Not found in the document.
17	Which component consumes the port Rp_AmbientLux?	Not found in the document.
18	What is the CAN ID of the ACU_Crash message?	Not found in the document.
19	What is the cycle time of the DoorLockStatus signal?	Section: 6. Functional Flow: Central Locking The central-locking flow has six steps: 1. A lock request arrives over CAN in the signal LockRe
22	What does the BCM do if the ambient light sensor is faulty?	Section: 9. Diagnostics and Error Handling The BCM supports these UDS services: 0x10 DiagnosticSessionControl, 0x22 ReadDataByIdentifier, 0x
28	What is the date of the approved version 1.0?	Not found in the document.

Full per-question results are in Evaluation_Results/results.csv and sample outputs with citations are in Evaluation_Results/sample_outputs.md.

7. Limitations

- One synthetic document of 14 pages; text and simple tables only, no diagrams and no OCR.
- A small local model can miss multi-part answers or cite the wrong page.
- The refusal threshold is set by hand and may block valid questions or allow weak matches.
- The refusal threshold (0.335) was tuned on the same 30 evaluation questions, so the refusal rate is optimistic.
- Keyword-based answer checking can mark a correct paraphrase as wrong.
- The system assists engineers; every answer must be checked against the cited page.

8. External dependencies declared

Internet is needed once to download the embedding model (Hugging Face) and the LLM (Ollama). At run time everything works locally. No cloud API is used.

9. Future work

Revision comparison, inconsistency checks, graph relationships between components, role-based access, OCR for scanned pages, and a FastAPI service layer.

10. Contribution

Sajjan Pawar: designed the HLD document and ground-truth set, implemented ingestion, retrieval, prompting, evaluation and the Streamlit interface, and wrote this report.

11. References

AUTOSAR Classic Platform concepts (public documentation); BAAI BGE embeddings; ChromaDB; Ollama; Streamlit; pdfplumber.